UNIT I – INTRODUCTION TO DATA MINING

Overview

- Data Mining is the process of extracting useful patterns from large datasets
- It is a key step in Knowledge Discovery in Databases (KDD)
- Used in business intelligence, healthcare, banking, and education analytics

Data Mining Tasks

- Classification – assigning data to predefined classes
- Clustering – grouping similar data
- Association – finding relationships among variables
- Regression – predicting continuous values
- Outlier detection – identifying anomalies

Architecture of Data Mining System

- Data Sources – databases, warehouses, web
- Preprocessing – cleaning and transformation
- Mining Engine – core algorithms
- Pattern Evaluation – identifying interesting patterns
- User Interface – visualization tools

Data Preprocessing

- Data Cleaning – handling missing and noisy data
- Data Integration – combining multiple sources
- Data Transformation – normalization and aggregation
- Data Reduction – dimensionality reduction

Issues in Data Mining

- Scalability and performance
- Handling high dimensional data
- Privacy and security concerns
- Integration with databases
- Interpretability of results

UNIT II – OVERVIEW OF DATA MINING ALGORITHMS

Models and Patterns

- Model is a representation of real-world process
- Patterns represent relationships in data
- Used for prediction and description

Scoring Functions

- Used to evaluate model performance
- Predictive accuracy measures
- Descriptive interestingness measures

Model Structures

- Regression models
- Classification models
- Clustering models
- Probabilistic models

Curse of Dimensionality

- High dimensional data leads to sparse data
- Distance measures become less meaningful
- Requires dimensionality reduction techniques

Model Evaluation

- Accuracy, Precision, Recall
- Cross-validation
- Overfitting and underfitting
- Robustness to noise

UNIT III – CLASSIFICATION

Decision Trees

- Tree structure with nodes and branches
- Algorithms: ID3, C4.5, CART
- Uses entropy and information gain

Bayesian Classification

- Based on Bayes theorem
- Naive Bayes assumes independence
- Efficient and simple

Rule-Based Classification

- IF-THEN rules
- Easy to interpret
- Generated from decision trees

Advanced Methods

- Neural Networks – backpropagation learning
- Support Vector Machines – margin maximization

- Frequent pattern based classification

Improving Accuracy

- Ensemble methods – bagging and boosting
- Pruning decision trees
- Feature selection

UNIT IV – CLUSTER ANALYSIS

Basic Concepts

- Clustering groups similar objects
- Unsupervised learning technique
- Used for segmentation and pattern discovery

Partitioning Methods

- K-means clustering
- K-medoids clustering
- Iterative refinement techniques

Hierarchical Methods

- Agglomerative approach
- Divisive approach
- Dendrogram representation

Density-Based Methods

- DBSCAN algorithm
- Identifies clusters of arbitrary shape
- Handles noise effectively

Advanced Clustering

- High dimensional clustering
- Graph clustering
- Constraint-based clustering

UNIT V – ASSOCIATION RULE MINING & VISUALIZATION

Association Rules

- Find relationships among items
- Example: Market basket analysis
- Rules: $X \rightarrow Y$

Frequent Itemsets

- Apriori algorithm
- Candidate generation
- Pruning techniques

Evaluation Metrics

- Support – frequency of occurrence
- Confidence – rule strength
- Lift – correlation measure

Advanced Techniques

- FP-Growth algorithm
- Parallel and distributed mining
- Incremental mining

Visualization

- Graphs and charts
- Multidimensional visualization
- WEKA tool usage